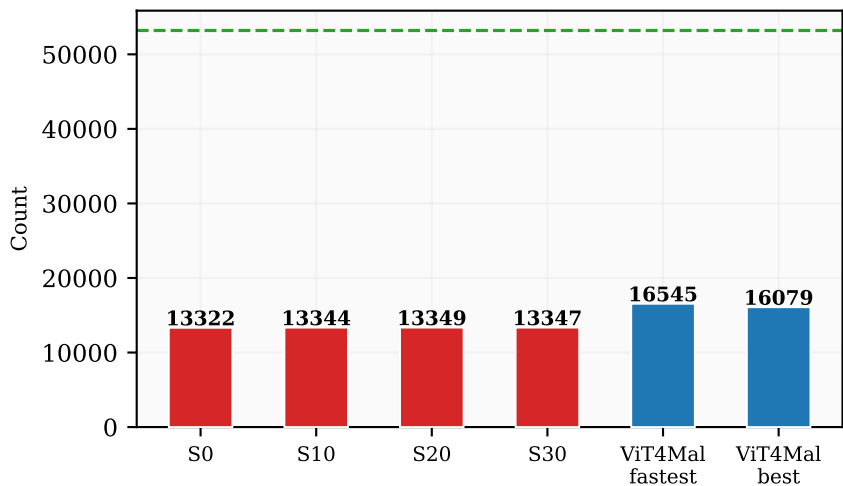
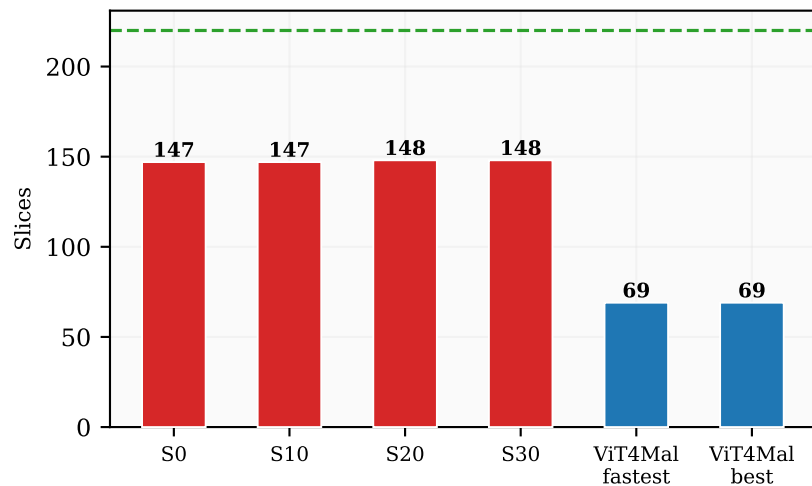


Resource Utilization — QKS-v2 vs ViT4Mal [2]

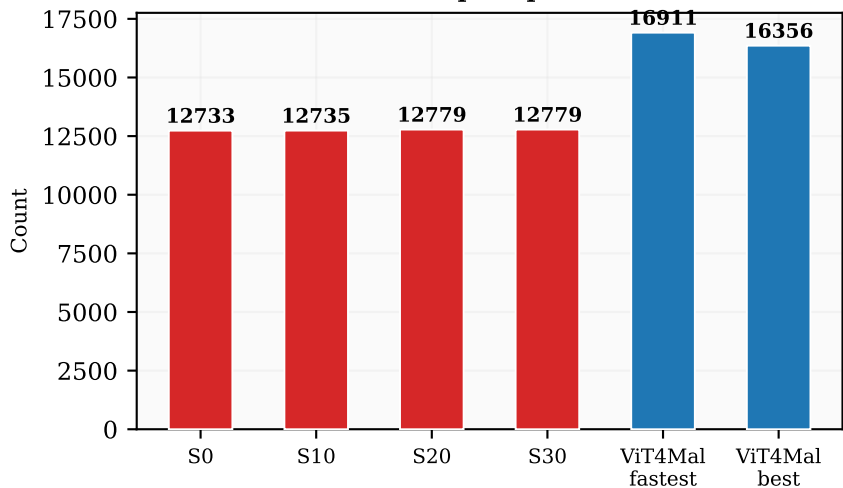
LUT



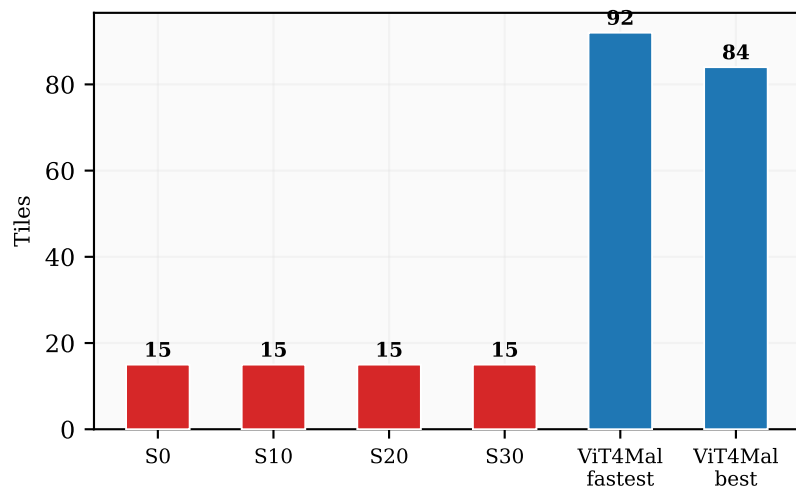
DSP



Flip-Flops



BRAM



QKS-v2: LUT ↓19%, FF ↓22%, BRAM ↓82% vs ViT4Mal. DSP ↑113% (parallelism tradeoff for <1.6 ms model-only inference).